VULLASA THRISHIKA

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Final-year B.Tech. Artificial Intelligence student at Amrita Vishwa Vidyapeetham with hands-on experience building AI/ML systems across Generative AI, NLP, computer vision, data analytics, and biomedical AI. Experienced in developing practical, research-oriented projects that combine machine learning, intelligent systems, and data-driven problem solving. Research experience at IIT (BHU) in biomedical signal processing, including EEG analysis, feature extraction, dataset analysis, and computational pipeline development.

EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore 2023 – 2027
B.Tech. in Artificial Intelligence | CGPA: 6.5/10

TECHNICAL SKILLS

Languages: Python, C, SQL, MATLAB
AI & Machine Learning: Machine Learning, Deep Learning, Generative AI, NLP, Computer Vision, Reinforcement Learning
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Transformers, LangChain, LangGraph
Data & Big Data: Apache Spark, PySpark, Hadoop, Spark SQL, MapReduce, Data Analysis
Research & Tools: Signal Processing, Bioinformatics, Git, GitHub, Docker, Linux

EXPERIENCE

Research Intern — Biomedical Signal Processing May 2026 – Jun 2026
Indian Institute of Technology (BHU), Varanasi

- Worked on biomedical signal processing involving EEG-related data, feature extraction, dataset analysis, and computational pipeline development.
- Applied signal processing and computational analysis techniques, supported by literature review and experimental analysis.

SELECTED PROJECTS

Social-to-Lead Agentic AI GitHub ↗

- Built a conversational AI system using RAG, intent classification, entity extraction, and LangGraph workflows for product assistance and lead qualification.
- Integrated FAISS and HuggingFace embeddings for product-specific retrieval and multi-turn conversations.

MathForge AI GitHub ↗

- Built a deterministic framework that converts natural-language math problems into algorithms, pseudocode, and Python code.
- Implemented rule-based semantic analysis, intermediate representations, algorithm identification, and complexity analysis.

NEXUS Energy Intelligence GitHub ↗

- Developed an energy analytics platform for household demand forecasting using ARIMA, LSTM, and hybrid ARIMA-LSTM models.
- Integrated SHAP explainability, anomaly detection, energy-system simulation, and V2G analysis.

NeuroFusion GitHub ↗

- Built a multimodal biomedical AI framework for emotion and fatigue analysis using EEG, fNIRS, ECG, and EMG.
- Applied signal preprocessing, feature engineering, and machine learning/deep-learning methods to physiological data.

CrowdVision AI GitHub ↗

- Designed a computer vision framework for crowd-risk prediction using person detection, crowd representation, anomaly analysis, and risk reasoning.
- Integrated YOLO, ResNet-18, ViT-MAE, Monte Carlo Tree Search, and Bayesian Networks for crowd-state analysis.

LEADERSHIP EXPERIENCE

Project Team Lead — Academic Projects

- Coordinated task allocation, timelines, documentation, and technical presentations across AI and IoT team projects.

CERTIFICATIONS

- GenAI Powered Data Analytics Job Simulation — Tata / Forage, 2026
- Introduction to Deep Learning — Infosys Springboard, 2026
- Introduction to Artificial Intelligence — Infosys Springboard, 2026
- Introduction to Data Science — Infosys Springboard, 2026